



Development of a new pressure drop correlation for open-cell foams based completely on theoretical grounds: Taking into account strut shape and geometric tortuosity

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HIGHLIGHTS

- A generalized pressure drop correlation for open-cell foams is developed.
- A new definition/relation for geometric tortuosity of open-cell foams is proposed.
- The influence of strut shape and geometric tortuosity on pressure drop is presented.
- The proposed correlation for pressure drop prediction shows very promising results.

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ABSTRACT

Owing to their outstanding geometry and material related properties, open-cell foams as structured packing in reactors, columns or heat exchangers offer clear advantages over their traditional counterparts. For the reliable and efficient design of such systems, an accurate knowledge or prediction of the pressure drop is of fundamental importance, as it plays an important economic role in situations involving high space velocities (high Reynolds numbers). The state-of-the-art pressure drop correlations for open-cell foams are not predictive as they tend to show large deviations from measured data. These deviations are observed mainly due to empirical coefficients in the correlations that have been fitted to a particular data set, but without theoretical meaning these coefficients cannot be extrapolated to other foam structures. In addition, the incorporation of inappropriate definitions (originating from misinterpretation or negligence of important aspects) of the structural and geometrical parameters of foams can also induce a considerable error. In this paper, using the notion of geometric tortuosity and starting from the basic Hagen–Poiseuille equation, a generalized pressure drop correlation for open-cell foams is developed which is based completely on theoretical grounds. The correlation is validated for a wide range of open porosities, pore sizes and materials and its applicability for different working fluids is demonstrated. Furthermore, the definitions of structural and geometrical parameters (as mentioned, these are crucial for pressure drop correlations but are sometimes misinterpreted or neglected) of open-cell foams are reviewed. A consensus on these definitions and their incorporation into the pressure drop correlations is proposed. Consequently, with the proposed correlation, pressure drop in open-cell foams can be predicted by knowing only two structural parameters: the open porosity and the window diameter.

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1. Introduction

Open-cell foams are highly porous reticulated cellular materials with a sponge-like structure (therefore sometimes also termed as

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sponges) that consists of an assembly of three dimensionally interconnected struts. Due to their high porosities and continuous three-dimensional cellular structure, they exhibit fascinating properties which can be beneficial in many industrial applications. In chemical engineering, the use of open-cell foams (ceramic or metal) as structured packing in reactors, columns or heat exchangers offer clear advantages over conventional randomly packed fixed beds. These advantages include high specific surface area, high mechanical strength, excellent thermal conductivity

Nomenclature

Symbols

A	coefficient of viscous term (–)
$(A)_{d_h}$	coefficient of viscous term, w.r.t hydraulic diameter (–)
$(A)_{d_p}$	coefficient of viscous term, w.r.t particle diameter (–)
a_p	free area perpendicular to the fluid flow (m^2)
B	coefficient of inertial term (–)
$(B)_{d_h}$	coefficient of inertial term, w.r.t hydraulic diameter (–)
$(B)_{d_p}$	coefficient of inertial term, w.r.t particle diameter (–)
d_c	cell diameter (m)
d_h	hydraulic diameter of open-cell foams (m)
D_h	hydraulic diameter of empty tube (m)
D_p	characteristic pore size ($d_w + d_s$) (m)
d_s	strut diameter (m)
d_w	window diameter (m)
f	drag coefficient (–)
k_1	Darcian permeability (m^2)
k_2	non-Darcian permeability (m)
L	length (m)
M	mass (g)
P	pressure (Pa)
$S_{v\text{-}geo}$	geometric specific surface area (w.r.t. geometric volume) (m^2/m^3)
$S_{v\text{-}solid}$	solid specific surface area (w.r.t. solid volume) (m^2/m^3)
V	velocity of the fluid (m/s)

Greek symbols

α	Ergun parameter for viscous term (–)
α^*	tortuous factor for viscous term (–)
β	Ergun parameter for inertial term (–)
β^*	tortuous factor for inertial term (–)
ΔP	pressure drop (Pa)
ε_n	nominal porosity (–)
ε_o	open porosity (–)
ε_s	strut porosity (–)
ε_t	total porosity ($\varepsilon_o + \varepsilon_s$) (–)
μ	dynamic viscosity of the fluid (Pa s)
v	volume (m^3)
ϕ	constant for shape of the strut cross-section (–)
ρ	density of the fluid (kg/m^3)
ρ_g	geometric density of foam (kg/m^3)
ρ_b	strut bulk density (kg/m^3)
ρ_s	solid density of foam (kg/m^3)
τ	geometric tortuosity of open-cell foams (–)

Abbreviations

CT	computed tomography
MRI	magnetic resonance imaging
MRPE	mean relative percentage error

(e.g. metal and SiC foams), high porosity and low resulting pressure drop. These benefits can be exploited especially in situations that involve high flowrates and/or strongly exothermic or endothermic processes [1–3]. In order to take advantage of these remarkable properties of open-cell foams, however, new reactor or column designs with open-cell foams as internals are inevitable [4]. In this context, for a reliable design an accurate knowledge or prediction of the pressure drop and/or heat transfer properties is a prerequisite [5–10].

Therefore, in the past decade, considerable research has been conducted on open-cell foams which has led to a greater understanding of their structure–property relationships and to the development of important correlations. Such correlations are necessary to predict the relevant data which in turn are required for designing the chemical engineering equipment such as, e.g., reactor, column and heat exchangers with open-cell foams as internals. However, recent literature shows that the state-of-the-art correlations for predicting the heat transfer and pressure drop properties of open-cell foams lack general applicability and therefore need improvement. Hence, further research is definitely needed in this area in order to come up with correlations featuring enhanced applicability. In this regard, the focus of the present work is to address the problem of pressure drop prediction in open-cell foams, thereby achieving a higher level of accuracy and reliability.

Today, in open literature a multitude of pressure drop correlations for open-cell foams can be found [1,11–20]. A study of recent literature reveals that these correlations, when applied to wider range of porosities or pore densities, tend to show a large deviation from the experimental data [3,5,18,19]. After reviewing the literature, the present authors are convinced that one of the main sources of deviation for most correlations is the use of empirical coefficients. Since these coefficients are usually determined for a limited set of data, they are not suitable for extrapolation to a wider range of pore densities, porosities, and/or working fluids,

thus causing a considerable deviation from measured pressure drop data.

Therefore, it is the aim of the present work to develop a general pressure drop correlation for open-cell foams completely based on theoretical grounds that allows for the pressure drop prediction solely based on easily measurable geometric foam properties and which thus bears an enhanced applicability and adaptability. That way, the use of any empirical coefficients or fitting of experimental pressure drop data can be avoided. For this purpose, we start with the basic Hagen–Poiseuille equation and derive a generalized pressure drop correlation for open-cell foams using the notion of geometric tortuosity. In order to determine the geometric tortuosity, we present a new correlation which is based on the geometric properties of open-cell foams and takes into account the shape of the strut cross-section.

In addition, in this paper a special emphasis is put on the structural and geometrical parameters of open-cell foams used in pressure drop correlations (which may be confused, misinterpreted or neglected) that directly or indirectly affect the pressure drop predictions and hence must be defined and determined accurately as well as applied correctly. These parameters include porosity (nominal, total or open), pore size (cell or window), shape of the strut cross-section (circular, triangular or concave triangular) and the specific surface area (based on solid volume “ $S_{v\text{-}solid}$ ” or based on geometric volume “ $S_{v\text{-}geo}$ ”). The incorporation of incorrect or inappropriate definitions of these parameters in pressure drop correlations can be another source of error resulting in significant deviation from experimental pressure drop data. For that reason, the definitions and the use of these parameters in literature pressure drop correlations is reviewed. A further aim is to propose a general consensus on the definitions and methods for the determination of these parameters and finally their incorporation into the pressure drop correlations in order to eliminate and/or avoid any confusion or misinterpretation for future research in this area.

2. Pressure drop modeling in open-cell foams

2.1. Theoretical background

In open literature, pressure drop measurement and modeling on open-cell foams has been a subject of intense investigation lately with the aim to correlate the flowrate with the pressure drop by taking into account the structural and geometrical parameters (e.g. porosity, pore diameter, strut diameter, geometric/solid specific surface area) of the open-cell foams. In most cases, Forchheimer-type resp. Ergun-type equations were applied in order to describe the pressure drop. This was achieved by determining the permeabilities for Forchheimer-type equations, or by using the coefficients for viscous (A) and inertial (B) terms for Ergun-type equations, which were empirically determined for the open-cell foams. Some authors [21,22] even used the numerical values of the coefficients (A and B) which were originally determined for packed beds of particles. Since Forchheimer- and Ergun-type equations represent the basis for literature pressure drop correlations for open-cell foams, it is appropriate to discuss them in detail along with the origin of their coefficients.

The Forchheimer equation (Eq. (1)) has been broadly applied to evaluate the permeability/pressure drop of both granular (packed beds) and cellular (open-cell foams) porous media.

$$\frac{\Delta P}{L} = \frac{\mu}{k_1} V + \frac{\rho}{k_2} V^2 \quad (1)$$

μ , ρ and V are the dynamic viscosity, density and velocity of the fluid, and k_1 and k_2 are the Darcian and non-Darcian permeabilities, respectively.

The most widely accepted interpretation of permeabilities in the Forchheimer equation was presented by Ergun and Orning in 1949 [23]. They proposed the following equation (Eq. (2)) which is also sometimes termed as the basic or original form of the Ergun equation [4,24,25]. Ergun and Orning obtained this form after substituting the hydraulic diameter d_h in terms of solid specific surface area $S_{v-solid}$ in their starting equation. Eq. (2) can also be applied using the geometric specific surface area which can be obtained by multiplying $S_{v-solid}$ by $(1 - \varepsilon_o)$.

$$\frac{\Delta P}{L} = \alpha \frac{S_{v-solid}^2 (1 - \varepsilon_o)^2}{\varepsilon_o^3} \mu V + \beta \frac{S_{v-solid} (1 - \varepsilon_o) \rho}{\varepsilon_o^3} V^2 \quad (2)$$

$\alpha = 2\alpha^*$ and $\beta = \beta^*/8$: Ergun parameters for viscous and inertial terms respectively

α^* and β^* : Tortuosity factors

ε_o : Open porosity (hydrodynamically relevant)

$S_{v-solid}$: Specific surface area w.r.t. solid volume

In order to introduce the particle diameter d_p in Eq. (2), Ergun [26] replaced $S_{v-solid}$ with the particle diameter d_p (Eq. (3)), based on the simple geometry of a sphere.

$$d_p = \frac{6}{S_{v-solid}} \quad (3)$$

He further introduced the values of $2\alpha^* = 4.17$ and $\beta^*/8 = 0.292$ in Eq. (2) that gave the following form (Eq. (4)) which is commonly known as Ergun equation for packed beds:

$$\frac{\Delta P}{L} = (A_{Ergun})_{d_p} \frac{(1 - \varepsilon_o)^2}{\varepsilon_o^3} \frac{\mu V}{d_p^2} + (B_{Ergun})_{d_p} \frac{(1 - \varepsilon_o)}{\varepsilon_o^3} \frac{\rho V^2}{d_p} \quad (4)$$

$(A_{Ergun})_{d_p}$ and $(B_{Ergun})_{d_p}$ are the coefficients for viscous and inertial terms (w.r.t. particle diameter), respectively, and are given as follows:

$$(A_{Ergun})_{d_p} = (\alpha)6^2 \Rightarrow (2\alpha^*)6^2 = 150 \quad (5)$$

$$(B_{Ergun})_{d_p} = (\beta)6 \Rightarrow \left(\frac{\beta^*}{8}\right)6 = 1.75 \quad (6)$$

According to the literature, the value of α^* [23,27,28] and β^* [23] represent the effects of the tortuous flow path of the fluid flow in a porous medium, which can be determined experimentally or derived theoretically. Ergun [26], however, preferred the empirical values of $\alpha = 4.17$ and $\beta = 0.292$, which already contain the tortuosity factors, i.e. α^* and β^* (see Eq. (2)). He determined these values by fitting a large amount of experimental pressure drop data of packed beds.

Eq. (4) is strictly applicable only to Newtonian flow in the experimental porosity range of approximately 0.35–0.5 [29]. This leads to the conclusion that the Ergun equation for packed beds with unchanged coefficients $(A_{Ergun})_{d_p}$ and $(B_{Ergun})_{d_p}$ must not be applied to open-cell foams that usually show much higher porosities than the upper applicability limit of 0.5.

Macdonald et al. [30] revisited the Ergun equation for packed beds and suggested $(A_{Macdonald})_{d_p} = 180$ and $(B_{Macdonald})_{d_p} = 1.8$ for Eq. (4).

In order to enhance the adaptability and generalization of the Ergun equation, Du Plessis et al. [29] derived theoretical correlations for the coefficients in Eq. (4). For the above-mentioned porosity range, they obtained average values of $(A_{Du Plessis})_{d_p} = 176$ and $(B_{Du Plessis})_{d_p} = 1.81$, thus confirming the range of empirical values of Ergun [26] and Macdonald et al. [30] (for packed beds) by their theoretical approach.

2.2. State-of-the-art pressure drop correlations for open-cell foams

Edouard et al. [5] presented a review on the state-of-the-art pressure drop correlations/models for open-cell foams and concluded:

“It appears from this study that no model is perfect and that the standard deviation between experimental and theoretical values can be as high as 100%.”

In recent years, some more studies were reported, e.g. [18–20], where literature data/correlations for pressure drop in open-cell foams were evaluated and new Ergun-type correlations or modifications of the existing ones were presented. The correlations from these studies are discussed in detail in the following paragraphs. It is interesting to note that in open literature, α and β , and hence A and B , are usually referred to as constants. However, as stated above (also clear from Eqs. (5) and (6)), the values of A and B depend upon the tortuosity factors α^* and β^* . These, in turn, should depend on the geometry of the porous medium, as it guides the fluid to a certain tortuous path. Hence, A and B cannot be regarded as constants.

In order to incorporate the geometric/structural properties of foams in the coefficients for viscous and inertial terms, Inayat et al. [20] proposed dimensionless semi-empirical correlations for α and β (from which A and B can be deduced) which are given as follows:

$$\alpha = \left[\left(\frac{\text{Window diameter}}{\text{Strut diameter}} \right) (\varepsilon_o) \right]^n \quad (7)$$

$$\beta = \left[\left(\frac{\text{Window diameter}}{\text{Strut diameter}} \right) (1 - \varepsilon_o) \right]^n \quad (8)$$

Though the use of these correlations gave close agreement to the experimental data, they still contained a fitting parameter (i.e. exponent n) that lacks a proper physical interpretation.

Recently, Dietrich [19] presented another Ergun-type semi-empirical pressure drop correlation for ceramic and metal foams which is given as follows:

$$\frac{\Delta P}{L} = (A_{\text{Dietrich}})_{d_h} \frac{\mu}{\varepsilon_o d_h^2} V + (B_{\text{Dietrich}})_{d_h} \frac{\rho}{\varepsilon_o^2 d_h} V^2 \quad (9)$$

$(A_{\text{Dietrich}})_{d_h} = 110$ and $(B_{\text{Dietrich}})_{d_h} = 1.45$ are coefficients w.r.t. hydraulic diameter and are empirically determined.

Note that the porosity in Eq. (9) is the open or hydrodynamic related porosity ε_o . However, Dietrich preferred to use the total porosity (ε_t) neglecting the strut porosity ε_s , here $\varepsilon_t = \varepsilon_o + \varepsilon_s$ (for details see Section 3.1). Since ε_s does not contribute to the pressure drop, as it is hardly accessible by the fluid, the use of ε_t might induce some error. Also, Dietrich [19] did not discuss the impact of the strut cross-section on the pressure drop. Depending upon the porosity, the strut cross-section may have different shapes (e.g. circular, triangular, or concave triangular (see Section 3.3.2)), which may have significant impact on $S_{v\text{-geo}}$ and consequently on the pressure drop.

Dietrich applied his correlation (using ε_t) to predict experimental pressure drop data from open literature and the root mean square deviation (RMSD) values for all predictions were scattered in the range of 34.5–87.5%. Dietrich argued that by taking into account the differences in the specific surface areas in some literature studies the overall RMSD value could be reduced. He finally concluded that the correlation gives predictions mostly within an error range of $\pm 40\%$ which represented an improvement as compared to the performance of previous pressure drop correlations existing in literature. This is the reason why the correlation of Dietrich is considered as benchmark and discussed in great detail in our present study.

Please note that Dietrich derived his correlation by using the definition of the friction factor for packed beds. The main difference between the Ergun equation for packed beds and the correlation from Dietrich is the different numerical value of the coefficients for the viscous and the inertial term. The second difference is that Ergun in his equation replaced the hydraulic diameter by the particle diameter, whereas Dietrich used the hydraulic diameter. Nevertheless, both used the same definition of the hydraulic diameter, so it is merely a different representation of the characteristic length.

More recently, Kumar and Topin [18] proposed a semi-empirical correction factor to be used in the correlation of Dietrich (Eq. (9)). The correction factor is given as follows:

$$C = \left(\frac{\text{Strut radius}}{\text{Strut length}} \right)^k \varepsilon_o \quad (10)$$

Here, k is a fitting parameter for which no physical interpretation is provided. In this regards, Eq. (10) seems to show a similar structure as compared to Eqs. (7) and (8). According to Kumar and Topin, the correction factor must be applied to coefficients determined according to Dietrich's method as follows:

$$(A_{\text{Kumar-Topin}})_{d_h} = (A_{\text{Dietrich}})_{d_h} \cdot C \quad (11)$$

$$(B_{\text{Kumar-Topin}})_{d_h} = (B_{\text{Dietrich}})_{d_h} \cdot C \quad (12)$$

Kumar and Topin determined $(A_{\text{Dietrich}})_{d_h}$ and $(B_{\text{Dietrich}})_{d_h}$ using Eq. (9), for which d_h was calculated with Eq. (16). However, in this case in order to determine $(A_{\text{Dietrich}})_{d_h}$ and $(B_{\text{Dietrich}})_{d_h}$, one still would need experimental pressure drop data for open-cell foam. This means that, to be able to apply Eqs. (11) and (12), the pressure drop of individual open-cell foam samples must be measured first.

Moreover, in order to calculate the Ergun coefficients (α and β) for the original form of the Ergun equation (Eq. (2)), Kumar and

Topin [18] have used an inappropriate definition of the specific surface area. According to the derivation from Ergun and Orning [23], the specific surface area to be used in Eq. (2) must be related to the solid volume $S_{v\text{-solid}}$. However, Kumar and Topin [18] used the specific surface area per unit geometric volume, i.e., $S_{v\text{-geo}}$ (which is referred to as a_c in their nomenclature). The difference between $S_{v\text{-geo}}$ and $S_{v\text{-solid}}$ is obvious from the following relation:

$$S_{v\text{-geo}} = S_{v\text{-solid}}(1 - \varepsilon_o) \quad (13)$$

This means that in the Ergun equation (Eq. (2) in [18]), either $S_{v\text{-geo}}$ (in their nomenclature a_c) must be replaced by $S_{v\text{-solid}}$, or the factors $(1 - \varepsilon_o)^2$ and $(1 - \varepsilon_o)$ must be removed from the numerators of the viscous and inertial term, respectively. From this explanation it seems that these authors confused the definition of geometric and solid specific surface area; hence the Ergun coefficients reported by them are questionable.

From the discussion in this section, it is clear that in open literature there exist quite some pressure drop correlations for open-cell foams, however, none of them exhibits a general applicability. Furthermore, the definitions of the structural and geometrical parameters of foams used in the pressure drop correlations are sometimes misinterpreted or neglected (strut porosity, shape of the strut cross-section, $S_{v\text{-solid}}$ and $S_{v\text{-geo}}$) which might be another source of deviation along with the usage of empirical coefficients. These issues are further discussed and addressed in the following sections of this paper.

2.3. Proposed pressure drop correlation for open-cell foams

Consider an empty tube of length L and inner diameter D . The pressure drop for laminar fluid flow through this tube can be given according to the Hagen–Poiseuille equation:

$$\frac{\Delta P}{L} = k_o 16 \frac{\mu}{D_h^2} V \quad (14)$$

Here, D_h is the hydraulic diameter ($4 \times$ cross-sectional area available for flow/wetted perimeter) and k_o is the shape factor that depends upon the cross-section of the tube. For higher flow velocities, a term for kinetic energy losses may be added to the above equation [23,31] to give:

$$\frac{\Delta P}{L} = k_o 16 \frac{\mu}{D_h^2} V + \frac{1}{2} \frac{\rho}{D_h} V^2 \quad (15)$$

Eq. (15) can be applied for the same tube packed with a porous medium (here open-cell foam of open porosity ε_o , Fig. 2) using Kozeny's assumption [27]. Under this assumption, the porous medium is regarded as a uniform bundle of tubes such that the total geometric specific surface area and the open porosity and hence the hydraulic diameter is equal to that of the porous medium (here, open-cell foam). The hydraulic diameter of the open-cell foam d_h is given by:

$$d_h = \frac{4\varepsilon_o}{S_{v\text{-geo}}} \quad (16)$$

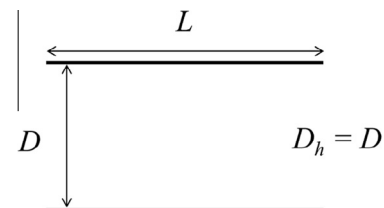


Fig. 1. An empty tube of length L and diameter D .

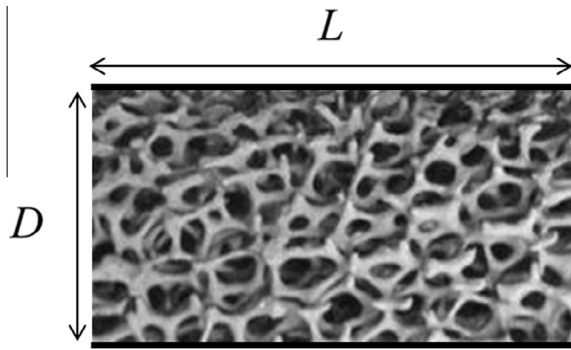


Fig. 2. Open-cell foam of porosity (ε_o) packed in a tube of length L and diameter D .

However, in this case the fluid has to flow through a tortuous path covering a longer (effective) distance L_e compared to the straight path L . Furthermore, the velocity is not going to be the same but higher. We shall call this velocity V^* .

Considering these facts, Eq. (15) can be translated for open-cell foams as follows:

$$\frac{\Delta P}{L_e} = k_o 16 \frac{\mu}{d_h^2} V^* + \frac{1}{2} \frac{\rho}{d_h} V^{*2} \quad (17)$$

The tortuous distance L_e depends upon the tortuosity of the flow (which in turn should depend upon the structural or geometrical properties of the porous medium packed in the tube) and can be given as follows [32–34]:

$$L_e = \tau L \quad (18)$$

Here, τ is the tortuosity, L is the straight path length of the tube and L_e the tortuous length. L_e can be defined as: $L_e = L + \text{additional path length due to the open-cell foam}$.

According to Carman [28] the velocity of a fluid through a porous medium of open porosity ε_o and tortuosity τ can be given as:

$$V^* = \frac{V}{\varepsilon_o} \tau \quad (19)$$

Taking the above definitions of velocity and tortuosity, Eq. (17) can be written as follows:

$$\frac{\Delta P}{L} = k_o 16 \tau^2 \frac{\mu}{\varepsilon_o d_h^2} V + \frac{\tau^3}{2} \frac{\rho}{\varepsilon_o^2 d_h} V^2 \quad (20)$$

Eq. (20) should give the pressure drop for fluid flow through a tube packed with porous media such as, e.g., an open-cell foam and is independent of any empirical coefficients. Substituting the value of the shape factor $k_o = 2$ for a circular tube [29], Eq. (20) gives the following form:

$$\frac{\Delta P}{L} = 32 \tau^2 \frac{\mu}{\varepsilon_o d_h^2} V + \frac{\tau^3}{2} \frac{\rho}{\varepsilon_o^2 d_h} V^2 \quad (21)$$

and

$$\frac{\Delta P}{L} = (A)_{d_h} \frac{\mu}{\varepsilon_o d_h^2} V + (B)_{d_h} \frac{\rho}{\varepsilon_o^2 d_h} V^2 \quad (22)$$

The coefficients of viscous A and inertia B terms (w.r.t. hydraulic diameter) are given as:

$$(A)_{d_h} = 32 \tau^2 \quad (23)$$

$$(B)_{d_h} = \frac{\tau^3}{2} \quad (24)$$

Please note that Eq. (22) is similar to the one derived by Dietrich (Eq. (9)). The only difference lies in the coefficients introduced.

Dietrich [19] used empirical values, whereas for the present work, A and B are derived on the basis of theoretical considerations and are functions of geometric properties of the open-cell foam. In order to determine the tortuosity in Eqs. (23) and (24), a new correlation (Eq. (35)) is derived in this paper and presented in the next section.

Using dimensionless numbers for velocity (Re: Reynolds number) and pressure drop (Hg: Hagen number)

$$\text{Re} = \frac{d_h V \rho}{\varepsilon_o \mu} \quad (25)$$

$$\text{Hg} = \frac{\Delta P}{L} \frac{d_h^3 \rho}{\mu^2} \quad (26)$$

the dimensionless form of Eq. (22) becomes:

$$\text{Hg} = (A)_{d_h} \text{Re} + (B)_{d_h} \text{Re}^2 \quad (27)$$

The proposed correlation (Eq. (22) or (27); with τ from Eq. (35)) is validated with experimental data covering a wide range of porosities, pore sizes, and materials of open-cell foams, and its applicability is demonstrated for different working fluids for a wide range of Reynolds numbers. The deviation from experimental data is given as the mean relative percentage error (MRPE) and is calculated for all predictions (see Section 4). In order to compare the different approaches of determining the coefficients $(A)_{d_h}$ and $(B)_{d_h}$ for open-cell foams, the values obtained from the literature correlations and the ones obtained from the present work are plotted vs. open porosity and are shown in Fig. 3.

From the results as shown in Fig. 3 it can be observed that the theoretical curves for $(A)_{d_h}$ and $(B)_{d_h}$ calculated by using the correlations given in Eqs. (23), (24) and (35) fall between the ones obtained from the literature. The opposite behavior for $(A)_{d_h}$ (from our former work [20]) is due to the empirical factor involved. The exponent in Eq. (7) (used to calculate $(A)_{d_h}$) was fitted to the experimental data of foams with porosities between 85% and 88% and hence cannot be generalized for a wider range of porosity. Nevertheless, the values obtained for $(A)_{d_h}$ for the above mentioned range of porosity using Eq. (7) are close to the ones from the present work. The performance of Eqs. (23) and (24) in calculating pressure drop (using Eq. (22)) is presented in Section 4 with calculated MRPE values.

2.4. Geometric tortuosity of open-cell foams

The concept of tortuosity for porous media was first introduced by Kozeny [27], who pointed out that for the fluid flow in a porous medium of length “ L ”, the effective (tortuous) path “ L_e ” covered by a fluid element is greater than L , which is due to the tortuous character of the flow path (Eq. (18)).

Sobieski et al. [33] pointed out that, when a fluid flows through a porous medium of length L , there are numerous (and not only one) possible channels. Each of these channels features a path length of L_i , thus giving a (microscopic) tortuosity of (L_i/L) . This will result in multiple (microscopic) tortuosities, and the overall tortuosity (L_e/L) will then be the average value of all microscopic tortuosities. This overall tortuosity is difficult to determine directly. However, one can consider an imaginary channel which has the same fluid conducting capacity as the sum of all microscopic channels. In the following paragraphs, a similar idea is used to derive a correlation for the overall tortuosity of open-cell foams.

According to the basic definition of tortuosity Eq. (18) can be written as follows:

$$\tau = \frac{L_e}{L} = \frac{L + (\text{additional length})}{L} \quad (28)$$

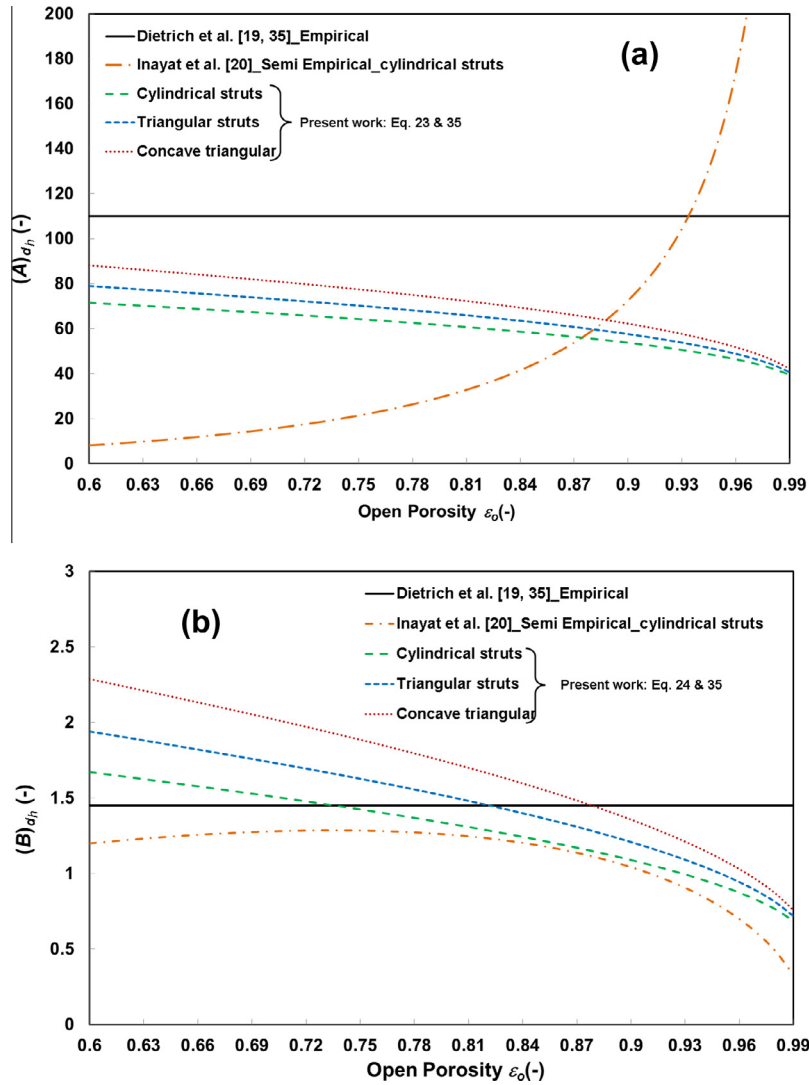


Fig. 3. A comparison of coefficients of viscous “ $(A)_{dh}$ ” (a) and inertial terms “ $(B)_{dh}$ ” (b).

Let us consider the same tube as shown in Fig. 1. The tube's geometric properties with respect to the internal diameter can be given as follows:

Surface area (S): πDL

Volume (v): $\pi D^2 L/4$

Specific surface area (S/v): $4/D$

Dimensionless specific surface area: $D(S/v) = 4$

Accordingly, the tortuosity of the empty tube is given by:

$$\tau_{tube} = \frac{L}{L} = 1 \quad (29)$$

Now we consider a cellular porous medium (open-cell foam) of open porosity ϵ_o packed in the same tube (Fig. 2). Then, the overall tortuosity (τ) can be written as follows (where L' is the additional length due to the open-cell foam):

$$\tau = \frac{L + L'}{L} = 1 + \frac{L'}{L} \quad (30)$$

By rearranging Eq. (30) one reaches the following solution:

$$\tau = 1 + \frac{\pi DL'/v}{\pi DL/v} = 1 + \frac{\pi DL'/v}{4/D} \Rightarrow \tau = 1 + \frac{D(\pi DL'/v)}{4} \quad (31)$$

Here, the additional path length L' is confined in the effective free volume of the open-cell foam (with a porosity of ϵ_o) which in turn

is placed into the tube with a total internal volume of v . Therefore, in Eq. (31), in order to account for the effective free volume, v is replaced by $v\epsilon_o$. Accordingly, the effective pore size will be reduced from the tube's internal diameter to the foam's average pore (window) size i.e. d_w . Eq. 31 can then be written as follows:

$$\tau \equiv 1 + \frac{d_w(\pi d_w L'/v)}{4\epsilon_o} \quad (32)$$

Let L' be the length of an imaginary tube whose surface area $\pi d_w L'$ is equal to the surface area of the open-cell foam, then the dimensionless geometric specific surface area w.r.t. total volume, i.e., $d_w(\pi d_w L'/v)$ should be equal to the dimensionless geometric specific surface area of the open-cell foam. Finally, Eq. (32) can be written as:

$$\tau \equiv 1 + \frac{\text{Dimensionless}(S_{v-geo})}{4\epsilon_o} \quad (33)$$

Here, S_{v-geo} is the geometric specific surface area of the open-cell foam. The above equation should give the overall tortuosity of a system consisting of a tube packed with an open-cell foam. We shall call this overall tortuosity “geometric tortuosity”, as it depends upon the geometric specific surface area of the open-cell foam.

Please note that the tortuosity as given in Eq. (33) represents the tortuosity of the fluid path inside the open-cell foam. For the

calculation of the tortuosity, we have considered an imaginary tube with the same fluid conducting capacity as that of the porous medium and whose specific surface area is equal to that of the porous medium skeleton. In other words, the additional length (L') for the fluid caused by the porous medium is dependent on the solid skeleton, but the calculation of the tortuosity is based on the usage of the inverse dimensionless hydraulic diameter (second term in Eq. (33)), which is related to the fluid phase.

S_{v-geo} and hence the dimensionless S_{v-geo} of open-cell foams can be determined experimentally by using volume image analysis such as, e.g., micro computed tomography (micro-CT) and magnetic resonance imaging (MRI) [3,35,36].

Alternatively, S_{v-geo} can be calculated by applying mathematical correlations using easily measurable geometric parameters of open-cell foams (e.g. window diameter, strut diameter, and open porosity). In open literature, there are several correlations available to calculate the specific surface area, though most of them tend to overestimate the values as compared to the measured literature data (for details see Section 3.3.1).

Considering the tetrakaidecahedron model, Inayat et al. [37] developed a theoretical correlation (Eq. (34)) for predicting the specific surface area of open-cell foams. This correlation gives close agreement to the measured data from open literature. The results as reported by Inayat et al. [37] were further confirmed by other authors [38–40].

$$S_{v-geo} = \phi \frac{[1 - 0.971(1 - \varepsilon_o)^{0.5}]}{d_w(1 - \varepsilon_o)^{0.5}} (1 - \varepsilon_o) \quad (34)$$

Here, d_w is the window diameter and ϕ is a geometric, non-empirical constant that depends upon the shape of the strut cross-section. The value of ϕ for cylindrical struts is ≈ 4.87 , for triangular struts ≈ 5.62 and for concave triangular struts ≈ 6.49 . The dimensionless S_{v-geo} (i.e., $d_w(S_{v-geo})$) from Eq. (34) is substituted in Eq. (33) to obtain the following equation:

$$\tau = 1 + \phi \frac{[1 - 0.971(1 - \varepsilon_o)^{0.5}]}{4\varepsilon_o(1 - \varepsilon_o)^{0.5}} (1 - \varepsilon_o) \quad (35)$$

In the literature, in order to calculate the tortuosity of cellular structures some authors [24,41,42] have presented mathematical correlations. These correlations are based on a cubic unit cell model and share a common definition (Eq. (36)) of the tortuosity as proposed by [41]:

$$\tau = \frac{L_e}{L} \equiv \frac{\text{Void volume}}{(\text{Min. free area} \perp \text{ to the flow})L} \equiv \frac{\varepsilon_o L^3}{a_p L} \quad (36)$$

Apparently, these correlations (unlike the present correlation (Eq. (35)) do not take into account the effect of the internal specific surface area of the tube in which the open-cell foam or cellular porous medium is packed. However, during pressure drop measurements and also in technical applications, the porous medium/open-cell foam is packed into a tube, and in this case the tortuous fluid flow should also depend upon the interplay of fluid elements between the structure of the porous medium and the tube wall/surface. Hence, the overall tortuosity is going to be different compared to the one obtained for an open-cell foam without external casing. Also, the tortuosity correlations based on Eq. (36) use cubic cell model and may not be suitable for open-cell foams, as it is already reported that the cubic cell model overestimates the specific surface area [36,37] and consequently the pressure drop [3] in open-cell foams. Besides, it is not clear how the minimum free area perpendicular to the flow direction (a_p) can be defined or even determined for irregular structures such as

open-cell foams. Hence, the use of cubic unit cell for pressure drop modeling in open-cell foams is not recommended. Nevertheless, for comparison purposes, the results from the literature correlations for tortuosity along with the ones obtained using Eq. (35) are shown in Fig. 4.

From these results it can be observed that the correlations from literature show higher values of tortuosity, in particular for the low porosity regime ($\varepsilon_o < 0.9$) compared to the present correlation. For higher porosities, the correlations show a similar trend. This means that the use of the correlations for tortuosity from literature in Eq. (22) will lead to an overestimation of the pressure drop in open-cell foams in the low porosity regime. This is in agreement with previous findings that the literature correlations based on the cubic cell unit model tend to overestimate the specific surface area and consequently the pressure drop in open-cell foams, especially at low porosities.

3. Geometric parameters of open-cell foams: Definitions and incorporation into pressure drop correlations

3.1. Porosity

Open-cell foams are manufactured in a wide range of porosity, typically between 0.7 and 0.95. Their porosity is mainly due to the void volume enclosed by the three dimensional strut network. However, an additional porosity (so-called “strut porosity”) may be present in the foam skeleton that originates from the manufacturing process (e.g., replication). Most commercial foams are manufactured by means of a replication technique [43] in which polymer foams are used as templates, and open-cell foams are obtained as positive images of the templates. As exemplarily shown in Fig. 5, this technique leads to hollow struts with internal void volume that stems from the burnt polymeric template [1,3,11]. Thus, the replication technique produces an extra void fraction, or strut porosity, which can be up to 5% [35] in addition to the open porosity of open-cell foams.

The internal void volume in the strut network may be avoided by using other manufacturing techniques such as, e.g., direct foaming (usually for ceramic foams [44]) or investment casting (usually for metal foams [45]).

In the literature, several types of porosities relevant for open-cell foams [1,3,35–37] are reported which can be summarized as follows:

- Open porosity ε_o or hydrodynamic porosity: relevant for fluid-dynamics.
- Strut porosity (ε_s , originates from the manufacturing process): irrelevant for fluid-dynamics [1,3,21].
- Total porosity ε_t : the sum of open and strut porosity.
- Nominal porosity ε_n : given by the manufacturer (usually no information is given regarding the open and strut porosity).

The total porosity ε_t can be determined by using the following equation:

$$\varepsilon_t = 1 - \frac{\rho_g}{\rho_s} \quad (37)$$

Here, ρ_g is the apparent density obtained from the apparent volume and mass of the foam sample, and ρ_s is the solid density which can be determined by He-pycnometry in case all pores are accessible. The open porosity ε_o can be calculated using Eq. (38).

$$\varepsilon_o = 1 - \frac{\rho_g}{\rho_b} \quad (38)$$

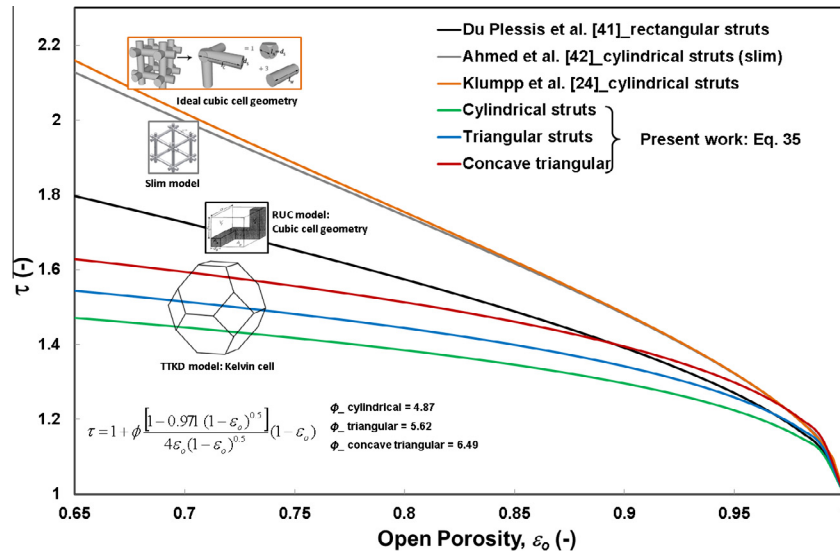


Fig. 4. Tortuosity vs. open porosity.

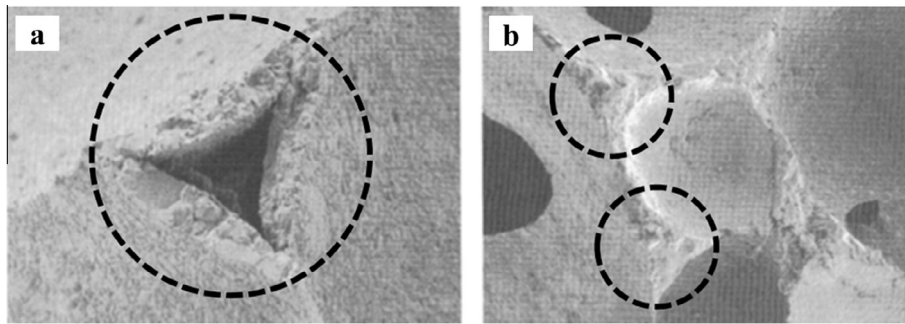


Fig. 5. Strut morphology: hollow struts from replication of polymer foams (a) and dense struts from direct foaming (b), adapted from [44].

Here, the strut bulk density ρ_b is given by Eq. (39), for which the true solid volume v_{solid} and hollow strut volume can be determined by He-pycnometry and mercury intrusion porosimetry respectively.

$$\rho_b = \frac{M_{foam}}{v_{solid} + v_{hollow-strut}} \quad (39)$$

According to the above paragraphs, the open porosity can be understood as the effective void fraction which is available for the fluid flow inside the open-cell foam structure. The strut porosity, in contrast, is the void fraction that does not play any role for the fluid flow because it is hardly accessible for the fluid [1,3]. Therefore, it can be concluded that the pressure drop in open-cell foams is a function of open porosity instead of total porosity. However, in open literature, still many authors tend to neglect the strut porosity and use the total porosity as a parameter in their pressure drop correlation, which may produce a significant error in the predictions. As an example, a comparison between the pressure drop prediction using open and total porosity is shown in Fig. 6. The data as well as the correlation is taken from [35].

The results as shown in Fig. 6 prove that neglecting the strut porosity is not acceptable. Even a strut porosity of ca. 4% can impart a significant impact on the pressure drop. Thus, it can be concluded that for pressure drop predictions in open-cell foams featuring hollow struts, the strut porosity should not be neglected. It is thus essential to use the open porosity ϵ_o (instead of total

porosity ϵ_t or nominal porosity ϵ_n) in the pressure drop correlations. The open porosity can be obtained by subtracting ϵ_s from ϵ_t .

For the present work, the external experimental data used for validating the correlation proposed in this paper were taken from those authors who have either given the open porosities or have investigated open-cell foams with no internal strut voids. Note that in case of open-cell foams without hollow strut volume (e.g. foams produced by direct foaming or investment casting), the strut porosity is essentially zero. Therefore, in that case open porosity can be considered as equal to the total porosity.

3.2. Pore size

The conventional way of determining the pore size in open-cell foams is counting the number of pores per linear inch (PPI). However, this definition of pore size is confusing due to the unclear definition of a “pore”, because a pore can be a cell or a window. Also, the PPI value (also referred to as “pore density” or “pore count”) used by the foam manufacturers does not give a precise measure, but shows a range of cell or pore sizes [11]. Usually, each foam manufacturer has its own reference scale, e.g., a foam considered as 80 PPI by one manufacturer could be defined as 110 PPI by another [44]. Nevertheless, the nominal PPI value is still used by some authors for calculating average pore size and as modeling parameter [46,47].

In the literature, different definitions of the characteristic pore size (other than PPI) have been used, which are given as follows:

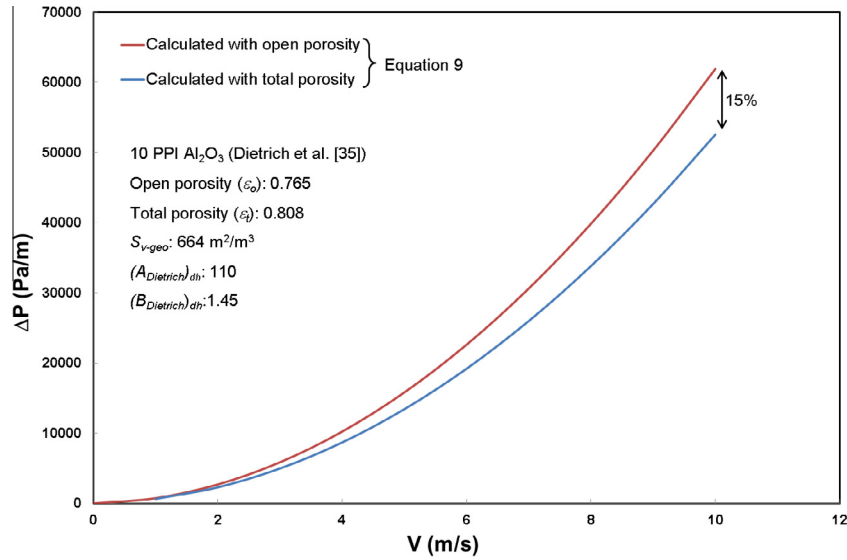


Fig. 6. Comparison of pressure drop: predicted by using ϵ_o (upper line) and ϵ_t (lower line). The data as well as the correlation is taken from [35].

d_w : average window diameter

d_c : average cell diameter

D_p : average value of the sum ($d_w + d_s$); where d_s is the strut diameter

In some studies, e.g. [21,22,42], d_w is calculated using Eq. (40) if only d_c is known. This method of calculating d_w is used by some foam manufacturers, and it is reported to give a good approximation [22,48].

$$d_w = \frac{d_c}{2.3} \quad (40)$$

From Fig. 7 it can be observed that a cell is an approximately spherical void volume enclosed by solid struts. A window, in contrast, is an opening into the cell and serves as a medium for cell-to-cell communication. When characterizing the pore size of open-cell foams it is very important to clearly indicate whether it is d_c , d_w , or D_p , as these three characteristic geometrical quantities might be confused with each other. For the sake of clarity, a sketch

or a picture giving the exact definition of the characteristic pore size should be provided [3,22]. Because the PPI value as given by the foam manufacturer is merely a nominal value indicating the range of cell or pore sizes, its use as a modeling parameter is not recommended.

For the present work the average window size (d_w) is used as the characteristic pore size for the open-cell foams. For the external data used as reference in this paper where the window size was not given, it is calculated using Eq. (40).

3.3. Geometric specific surface area (S_{v-geo})

Open-cell foams exhibit high geometric specific surface areas. It is one of the most important property of open-cell foams, as it is relevant for momentum, mass and heat transfer [1,3,36,49].

For pressure drop modeling in open-cell foams, S_{v-geo} or $S_{v-solid}$ (Eq. (13)) is used as modeling parameter, either directly (e.g. [11,25]) or indirectly (e.g. for computing the hydraulic diameter

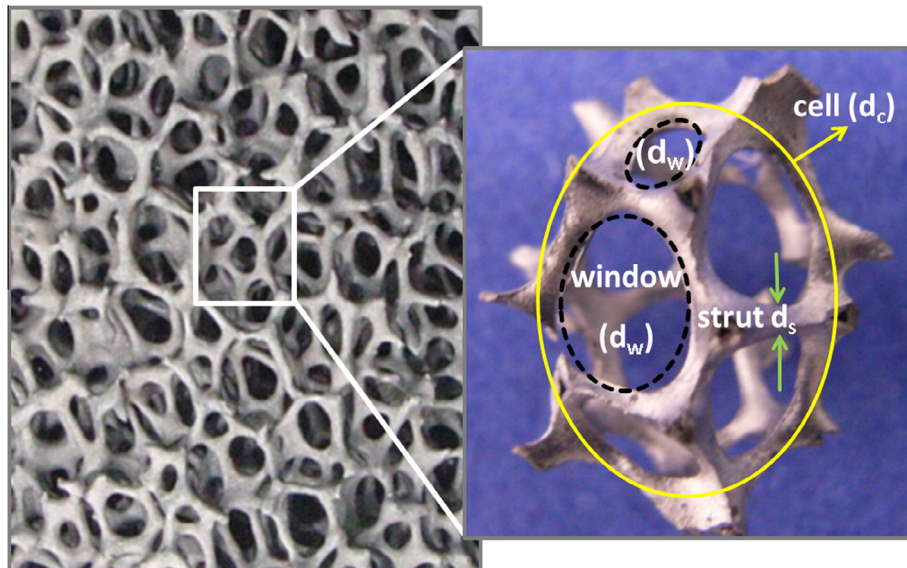


Fig. 7. Cell unit of open-cell foam showing its components.

or equivalent particle diameter [22,35]). Therefore, its accurate determination and incorporation (S_{v-geo} and $S_{v-solid}$ must not be confused with each other) into the pressure drop correlations is crucial.

The geometric specific surface area of open-cell foams is usually defined as the total external surface area of the struts per unit geometric volume, assuming a smooth strut surface [3]. In reality, the struts may exhibit a rough surface (especially in the case of ceramic foams) and possess a void volume (strut porosity, see also Section 3.1). Therefore, conventional physisorption methods such as, e.g., the BET method cannot be used to measure the external specific surface area experimentally, because these methods would give an overestimation. Instead of physisorption methods, volume image analysis such as, e.g., micro computed tomography (micro-CT) or magnetic resonance imaging (MRI) can be applied [3,35–37,39] to determine S_{v-geo} . However, these methods are time consuming, expensive and require quite some computational effort (especially regarding the post processing of the data) [24]. Therefore, mathematical correlations enabling the prediction of S_{v-geo} using properties that are easy to determine (e.g. window diameter, strut diameter, porosity) are of great interest. Such correlations are

usually developed by considering a geometric model based on a certain cell unit. A summary and evaluation of literature correlations for S_{v-geo} is presented in the next section.

3.3.1. Correlation for predicting S_{v-geo}

In open literature, several correlations for the prediction of S_{v-geo} of open-cell foams [1,21,22,36,37,50] can be found. In order to compare the performance of these correlations, the theoretical dimensionless form of S_{v-geo} is plotted versus the open porosity and compared with experimental data (Figs. 8a and b). Note that for this comparison only theoretical correlations are considered.

Fig. 8a shows that for the low porosity regime ($\varepsilon_o < 90\%$) almost all correlations share a common behavior of overestimating the experimental values. In addition, the correlations tend to produce widely differing results. For the high porosity regime ($\varepsilon_o > 90\%$), due to limited experimental data, a direct comparison between theoretical and measured values was not possible. However, in this regime most correlations produce quite similar results. In our previous contribution [37], we presented a correlation for estimating S_{v-geo} (Eq. (34)). The results obtained by using Eq. (34) in comparison with the experimental data are presented in Fig. 8b.

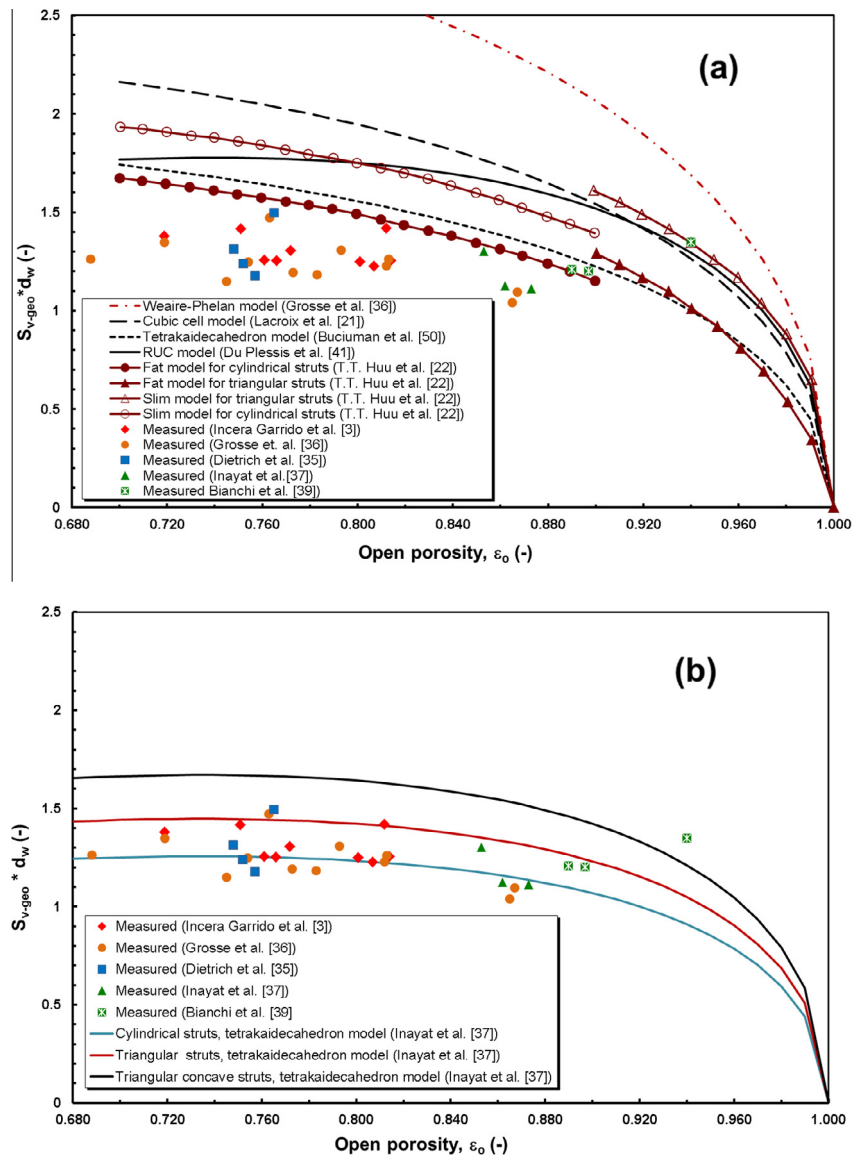


Fig. 8. Dimensionless form of S_{v-geo} vs. open porosity of open-cell foams: theoretical and measured values.

From Figs. 8a and b it can be observed that among the literature correlations the best results are produced by the correlation from Inayat et al. [37]. Please note that this correlation was derived on the basis of the tetrakaidecahedron model (Kelvin cell geometry) taking different shapes of the strut cross-section, namely cylindrical, triangular, and concave triangular, into account. The correlation makes use of d_w and ε_o as measured properties and predicts S_{v-geo} satisfactorily.

Thus, for a theoretical determination of S_{v-geo} of open-cell foams, we recommend using Eq. (34). For other literature correlations, we identified possible sources for the deviation, which are discussed in the following section.

3.3.2. Predicting S_{v-geo} : Possible sources of deviation

According to our observation, the large deviation of predicted values from the measured data (Fig. 8a) might be induced by two main factors, namely the selection of an inappropriate geometric model for the unit cell and the negligence of the shape of the strut cross section.

3.3.2.1. Selection of geometric model. From Figs. 8a and b it is clear that the selection of an appropriate geometric model for the description of the unit cell is very critical, as different geometric models will lead to different predictions even when using the same measured data (such as, e.g., pore size, strut size, or porosity). In current literature it is agreed that the Kelvin cell geometry is the most suitable representative geometry for open-cell foams [3,11,50,51]. This is also clear from the comparison presented in Fig. 8 a and b. Here, the correlation based on the Kelvin cell geometry gives the closest agreement to the experimental data as compared to the other model geometries (i.e., cubic, cubic RUC, or dodecahedron). In this respect we recommend using the tetrakaidecahedron geometry (so called “Kelvin cell”) as representative geometry for geometric and pressure drop modeling in open-cell foams. Nevertheless, in order to moderate the conclusion about the representative geometry, a detailed study on open-cell

foams to determine the percentage of cells which are topologically speaking consistent with the Kelvin cell, e.g. by applying X-ray micro computed tomography, would be of interest for future studies.

3.3.2.2. Shape of the strut cross-section. According to literature, depending on porosity, open-cell foams exhibit different shapes of their strut cross-section namely circular, triangular and concave triangular (see also Fig. 9) [22,52]. These variations in the shape of the strut cross-section can have a high impact on the global geometric specific surface area, as the surface to volume ratio in each case will be different. Thus, a correlation for predicting S_{v-geo} developed considering cylindrical struts (i.e., circular strut cross-section) must not be applied for foams with triangular or concave triangular struts, and vice versa. As a consequence, for theoretical estimation of S_{v-geo} by use of a mathematical correlation, the development and use of different forms of the correlation corresponding to each of the individual strut shapes is mandatory.

However, in literature only few authors have discussed the significance of considering strut variations in the geometric modeling of open-cell foams. The phenomenon of strut variation with porosity was first discussed by Bhattacharya et al. [52] in detail for metal foams. From the microscopic images, they observed that the strut cross-section changes from circular to triangular as the porosity reaches approximately 0.93. At even higher porosities the strut cross-section tends to get concave triangular.

Huu et al. [22] discussed only two types of strut cross-section shape, namely cylindrical and triangular. According to them, the porosity value at which the strut cross-section changes from circular to triangular is 90%. They postulated this assumption to be applicable to all solid foams including ceramic and metal. However, no evidence was given by Huu et al. for this assumption, while Bhattacharaya et al. gave evidence for metal foams.

Inayat et al. [37] concluded on the basis of own and external data that for open porosities lower than 90%, ceramic foams feature

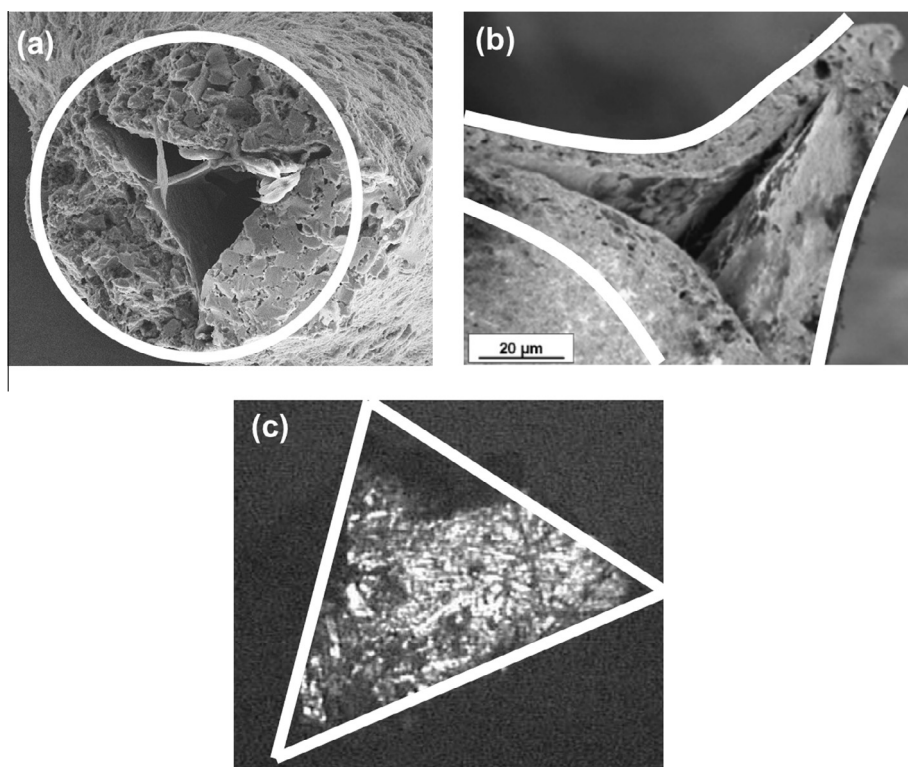


Fig. 9. Variation of the strut cross-section in open-cell foams: adapted from, (a) [37] SSiC foam, (b) [21] β -SiC foam, (c) [52] aluminum foam.

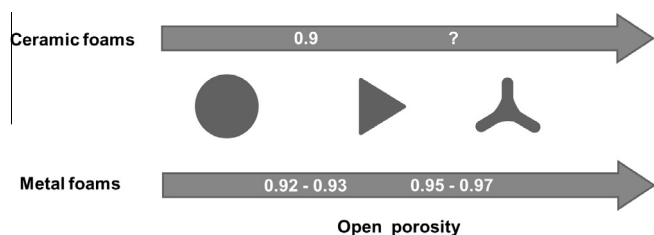


Fig. 10. Variation of the strut cross-section shape in open-cell foams: proposed borders based on information from [22,37,41,52], depending upon the manufacturing process, the struts may have internal void volume.

a circular strut cross-section, whereas for open porosities higher than 90% the struts tend to get triangular, and finally concave triangular.

Based on the studies mentioned in the above paragraphs, we propose porosity borders at which the strut cross-section changes its shape individually for ceramic and metal foams as given in

Fig. 10. For ceramic foams, due to insufficient information available in literature, the porosity border between triangular and concave triangular struts could not be identified.

It should be noted that the above-mentioned conclusions from different authors are still based on a limited set of experimental data. Thus, the porosity borders at which strut shapes change from circular to triangular or concave triangular may still vary for foams of different materials. Therefore, we recommend to closely observe the strut cross-section for each study meant for characterizing open-cell foams. When handling literature data where the strut cross-section is unknown, the assumptions as shown in **Fig. 10** can be applied as guideline. In order to obtain the missing information regarding the porosity border of ceramic foams and/or to relax these assumptions further data would be required.

4. Validation of the proposed pressure drop correlation

In order to demonstrate the applicability of the pressure drop correlation proposed in this paper, measured data from open

Table 1

Properties of open-cell foams (ceramic and metal).

Reference	Foam material	Fluid	PPI (–)	d_c (mm) (Measured)	d_w (mm) (Measured or Cal. $d_w = d_c/2.3$)	ε_t (–) (Measured)	ε_o (–) (Measured or as suggested by the authors)	$S_{v-solid}$ (m ^{–1}) (measured)	S_{v-geo} (m ^{–1}) Cal. by Eq. (34) or measured)
[41]	Aluminum	Water	45			0.978	0.978	186E3 ^a	4092 ^b
			60			0.975	0.975	252E3 ^a	6300 ^b
			100			0.973	0.973	440E3 ^a	11880 ^b
		Glycerol	45			0.978	0.978	186E3 ^a	4092 ^b
			60			0.975	0.975	252E3 ^a	6300 ^b
			100			0.973	0.973	440E3 ^a	11880 ^b
[54]	Aluminum	Air	5	4.221	1.835	0.8990	0.8990	–	583
			5	4.347	1.890	0.9300	0.9300	–	585
			10	3.422	1.488	0.9085	0.9085	–	699
			10	3.413	1.484	0.9386	0.9386	–	713
			20	2.784	1.210	0.9200	0.9200	–	953
			20	2.723	1.184	0.9353	0.9353	–	909
			40	1.935	0.841	0.9091	0.9091	–	1234
			40	1.727	0.751	0.9586	0.9586	–	1411
[53]	Aluminum	Water	10	6.900	3.000	0.921	0.921	–	383
			20	3.600	1.565	0.920	0.920	–	737
			40	2.300	1.000	0.928	0.928	–	1115
[1]	Aluminum	Air		4.700	2.044		0.927	–	548
				2.200	0.957		0.938	–	1281
				2.000	0.870		0.937	–	1417
[21]	β -SiC	AIR		3.650–3.700	1.598		0.914	–	852
				2.650–2.700	1.163		0.910	–	1186
				1.750–1.800	0.772		0.915	–	1757
[35]	Alumina	Air	10		2.253	0.808	0.765	–	664 ^c
			20		1.529	0.754	0.690	–	1090 ^c
			20		1.091	0.802	0.748	–	1204 ^c
			20		1.464	0.854	0.811	–	1109 ^c
			30		0.884	0.806	0.752	–	1402 ^c
			45		0.625	0.809	0.757	–	1884 ^c
	Mullite	Air	10		2.111	0.785	0.735 ^d	–	594
			20		1.348	0.736	0.686 ^d	–	922
			20		1.405	0.789	0.741	–	892
			20		1.522	0.834	0.784 ^d	–	816
			30		1.127	0.793	0.748	–	111
			45		0.685	0.797	0.744	–	1829
[20]	SSiC	Air	10		1.800	0.878	0.853	–	651
			20		1.297	0.896	0.873	–	875
			30		1.030	0.885	0.862	–	1122

In case of solid struts (no internal void volume), $\varepsilon_t = \varepsilon_o$

^a Experimental data [55].

^b $S_{v-geo} = S_{v-solid} (1 - \varepsilon_o)$.

^c Experimental data [35].

^d A strut porosity of 5% (as suggested [35]) is subtracted from ε_t to obtain ε_o .

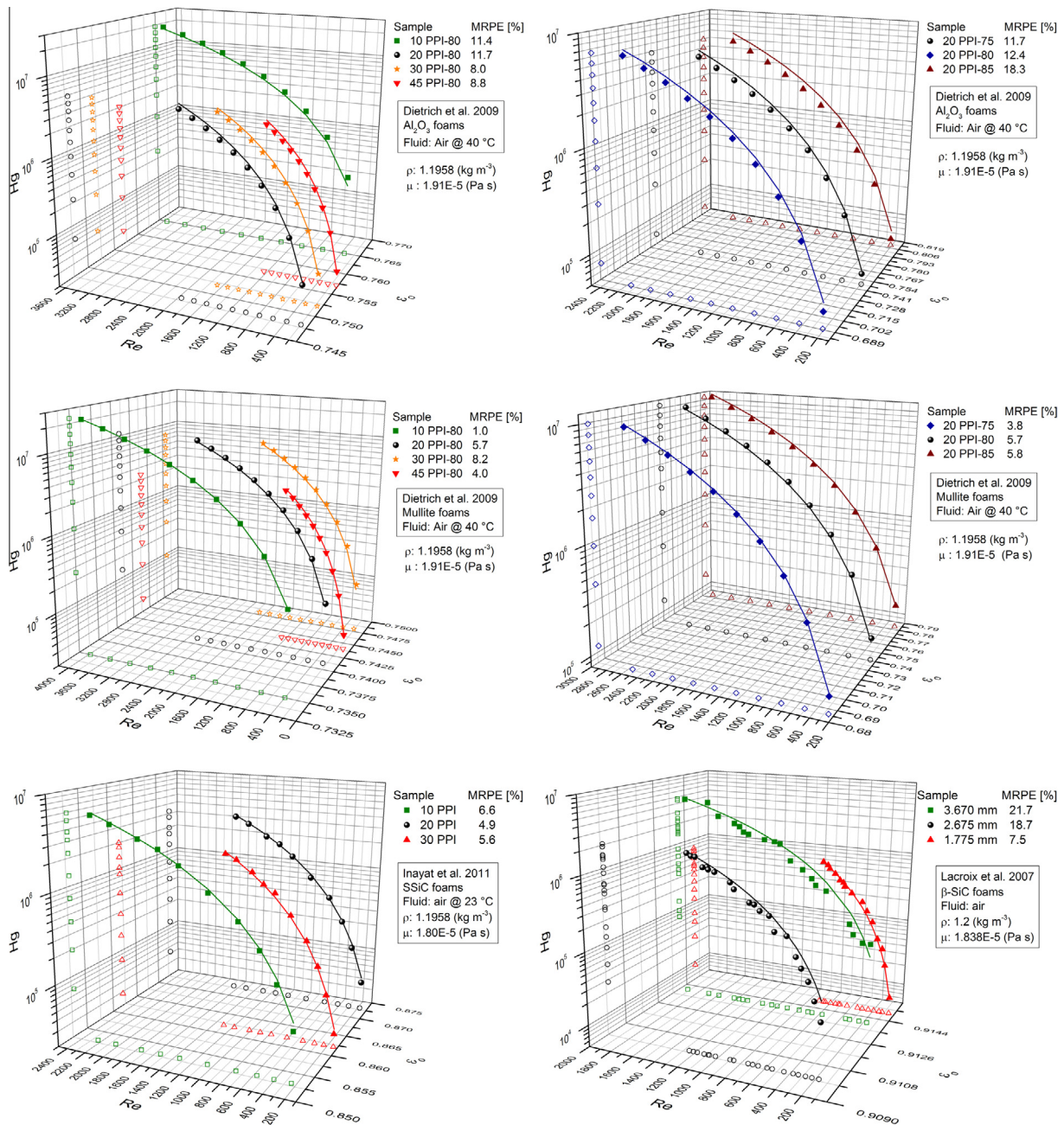


Fig. 11. Pressure drop prediction in ceramic foams: symbols (experimental data from Dietrich et al. [35], Inayat et al. [20], Lacroix et al. [21]), lines (correlation proposed in the present work).

literature [1,20,21,35,41,53–55] of foams of different porosities, pore sizes, and materials, obtained for different working fluids were used (Table 1). For the reproduction of the measured pressure drop data from literature, permeabilities (k_1 and k_2 ; for Eq. (1)) and drag coefficients f with k_1 (Eq. (41)) from the corresponding authors were used.

$$\frac{\Delta P}{L} = \frac{\mu}{k_1} V + \frac{\rho f}{\sqrt{k_1}} V^2 \quad (41)$$

where this information was not given, pressure drop data points were mapped (from the original pressure drop vs. velocity diagrams) using the digitizer module of the software *Origin 8.6G*.

In order to predict the pressure drop using the present correlation (Eq. (22) or (27)), one out of two sets of measured data is necessary. These data sets are given as follows:

1. ε_0 and d_w
2. ε_0 and S_{v-geo}

The specific surface area S_{v-geo} , when not given as measured property, was calculated using ε_0 and d_w using Eq. (34). In case d_w was not given, as some authors measured the cell diameter d_c only, d_w was approximated using Eq. (40). In addition, to be able to use the correct form of the correlation for S_{v-geo} , and consequently for geometric tortuosity and pressure drop, information on the shape of the strut cross-section is necessary. Where the shape of the strut cross-section was not given/shown in the respective literature source, it was assumed based on the proposed ranges shown in Fig. 10.

The deviation of the predictions from the measured data was determined using the mean relative percentage error (MRPE) by

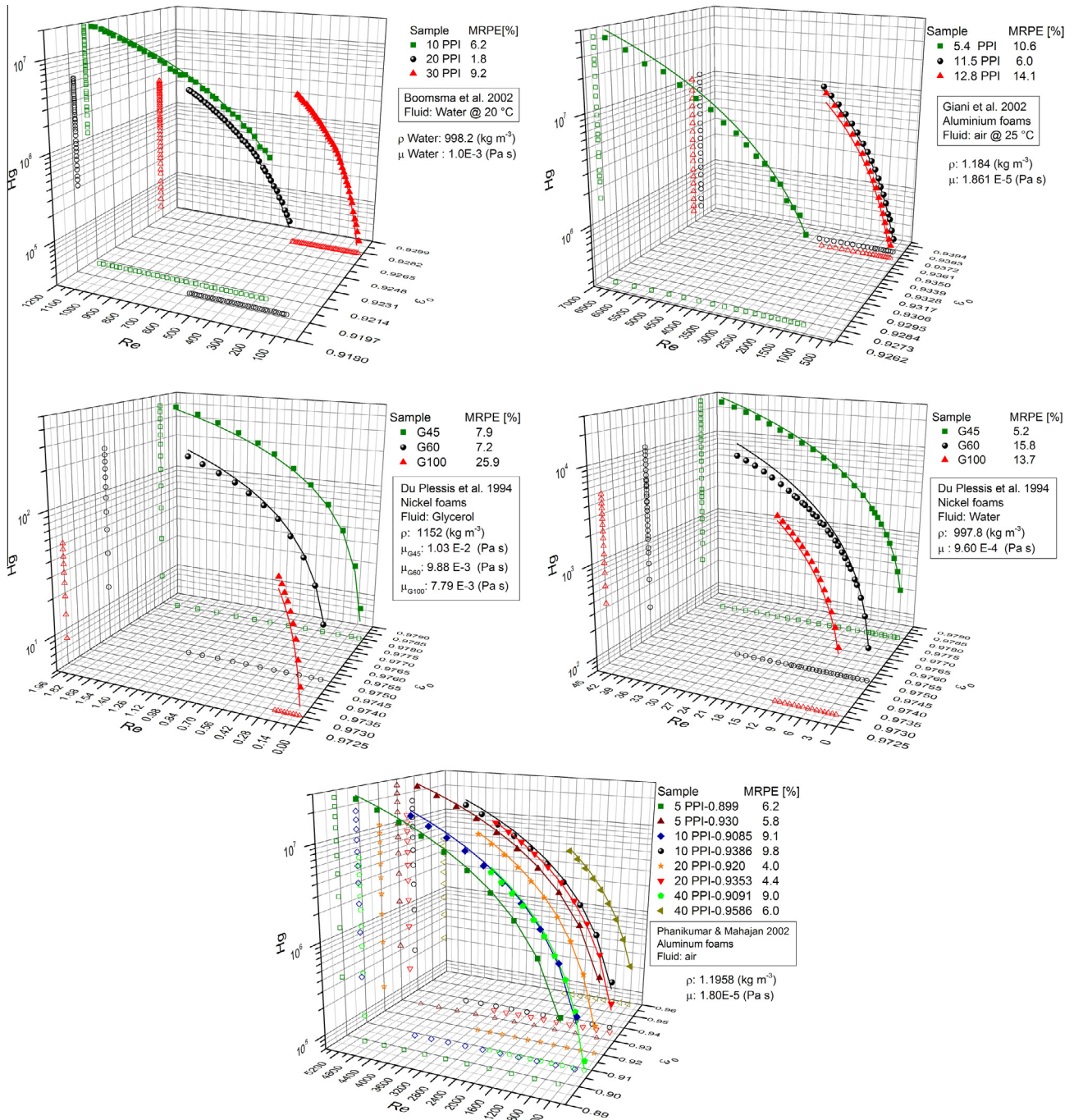


Fig. 12. Pressure drop prediction in metal foams: symbols (experimental data from Boomsma et al. [53], Giani et al. [1], Du Plessis et al. [41], Phanikumar and Mahajan [54]), lines (correlation proposed in the present work).

applying Eq. (42). The Hg -number, for both measured and calculated pressure drop, was determined using Eq. (26).

$$MRPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{(Hg_{\text{calculated}})_i - (Hg_{\text{measured}})_i}{(Hg_{\text{calculated}})_i} \right| \quad (42)$$

The comparison between measured and predicted pressure drop is shown in Fig. 11 (for ceramic foams) and Fig. 12 (for metal foams). From these results, it can be observed that the pressure drop in open-cell foams (ceramic or metal) can be predicted to a very satisfactory level with most predictions having a MRPE value of less than 10%. Thus, it can be concluded that a reliable pressure drop prediction in open-cell foams is possible by knowledge of

only two measured parameters, either ε_0 and d_w or ε_0 and S_{p-geo} . In this scenario, however, in order to relax the assumption of the porosity borders (necessary for the use of the correct form of the correlation for S_{p-geo}) for different strut shapes, more data is necessary. Therefore, the present authors recommend to always give the exact shape of the strut cross-section for future studies in this area.

5. Conclusions

A universal correlation for pressure drop prediction in open-cell foams is proposed (Eq. 22) which is solely based on theoretical

grounds and takes into account the shape of the strut cross-section and geometric tortuosity. For determining the geometric tortuosity of open-cell foams, a new relation is presented. The proposed pressure drop correlation is validated for a wide range of open porosities, pore sizes, and materials of open-cell foams and its applicability for different working fluids is verified. It is demonstrated that the pressure drop prediction in open-cell foams is possible by knowing only two measured structural parameters, i.e., window diameter and open porosity, which are easy to determine.

The definitions of structural and geometrical parameters of open-cell foams (which may be confused, misinterpreted or neglected) are reviewed. A consensus on these definitions and their incorporation into pressure drop correlations is proposed.

Open-cell foams exhibit different shapes of their strut cross-sections at different porosities. This fact can have a considerable influence on their geometric specific surface area and consequently on the pressure drop. Therefore, for geometric modeling of open-cell foams, the shape of the strut cross-section must not be neglected. It is shown that the correlation based on the tetrakaidecahedron model (Kelvin cell geometry), established for different strut shapes, predicts the geometric specific surface area most satisfactorily. Furthermore, the use of the geometric specific surface area (obtained from this correlation) as input for the calculation of the geometric tortuosity, and consequently for the pressure drop, shows very promising results.

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